DIP-SMC-PSO: Technical Progress Memo

Model choices, algorithm design, and experimental results

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Contents

1 System Model: Equations and Parameter Choices

1.1 State Vector and Coordinate Definition

The DIP state is a 6-vector:

$$\boldsymbol{x} = \left[x,\ \theta_1,\ \theta_2,\ \dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2\right]^\top \in \mathbb{R}^6 \qquad (1)$$

where x is cart position, θ_1 and θ_2 are pendulum link angles (from vertical), and dots denote time derivatives.

Coordinate convention and units. The generalized coordinates are

$$\boldsymbol{q} = [x,\ \theta_1,\ \theta_2]^\top, \qquad \dot{\boldsymbol{q}} = [\dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2]^\top \qquad (2)$$

with SI units: x in m, θ_i in rad, $\dot{x}$ in m/s, and $\dot{\theta}_i$ in rad/s. Positive x is rightward cart motion, and positive θ_i is counterclockwise from the upright vertical axis ($\theta_i = 0$).

Target equilibrium and error coordinates. The nominal stabilization target is the upright equilibrium:

$$\boldsymbol{x}^\star = [0,\ 0,\ 0,\ 0,\ 0,\ 0]^\top, \qquad \mathbf{e} = \boldsymbol{x} - \boldsymbol{x}^\star \qquad (3)$$

which is used in controller design and Lyapunov analysis. For tracking tasks, a nonzero cart reference updates only the first element of $\boldsymbol{x}^\star$.

Measured output and operational bounds. The feedback vector is

$$\boldsymbol{y} = [x,\ \theta_1,\ \theta_2,\ \dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2]^\top \qquad (4)$$

(full-state feedback in simulation). In HIL deployment, the same channels are obtained from encoder/estimator pipelines. Operational bounds used for pass/fail decisions are enforced by the benchmark configuration files.

1.2 Equations of Motion

The full Lagrangian model is:

$$\mathbf{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} + \mathbf{G}(\boldsymbol{q}) + \boldsymbol{F}_{\text{fric}} = \boldsymbol{B}\,u \qquad (5)$$

where $\boldsymbol{q} = [x,\theta_1,\theta_2]^\top$, $u \in [-150, 150]$ N is the cart force, and $\boldsymbol{B} = [1,0,0]^\top$ selects the actuated direction.

Modeling assumptions. The model assumes planar rigid-body dynamics, ideal revolute joints, bounded single-input actuation, and friction represented by viscous plus Coulomb terms. External disturbances and uncertainty scenarios are injected through benchmark profiles used in MT-8/LT-6 studies.

Equivalent first-order form. For control synthesis, (5) is written as

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + g(\boldsymbol{x})\,u + \boldsymbol{d}(t) \qquad (6)$$

where $\boldsymbol{d}(t)$ is zero in nominal simulations and nonzero in disturbance rejection experiments.

Structural properties used in stability analysis. For admissible configurations, $\mathbf{M}(\boldsymbol{q})(\boldsymbol{q})$ is symmetric positive definite, and the standard manipulator identity $\dot{\mathbf{M}}(\boldsymbol{q}) - 2\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})$ is skew-symmetric. These properties are used in Lyapunov-based controller proofs.

Model partition used in this workflow.

- **Full nonlinear model:** Used for reported benchmark metrics (MT-5, MT-7, MT-8, LT-6), including full coupling, Coriolis, centrifugal, gyroscopic, and friction effects.
- **Simplified model:** Used for PSO search loops with reduced coupling and omitted gyroscopic term to reduce optimization runtime.

Inertia matrix $\mathbf{M}(\boldsymbol{q})$ (3×3, symmetric, configuration-dependent): To keep the matrix compact in print, define $c_1 = \cos\theta_1$, $c_2 = \cos\theta_2$, and $c_{12} = \cos(\theta_1 - \theta_2)$.

$$\mathbf{M}(\boldsymbol{q}) = \begin{bmatrix} M + m_1 + m_2 & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \qquad (7)$$

with

$$\begin{aligned} a_{12} &= (m_1 l_{c1} + m_2 l_1)c_1 + m_2 l_{c2} c_2 \\ a_{13} &= m_2 l_{c2} c_2 \\ a_{22} &= m_1 l_{c1}^2 + m_2(l_1^2 + l_{c2}^2) + I_1 + I_2 + 2 m_2 l_1 l_{c2} c_{12} \\ a_{23} &= m_2 l_{c2}^2 + I_2 + m_2 l_1 l_{c2} c_{12} \\ a_{33} &= m_2 l_{c2}^2 + I_2 \end{aligned} \qquad (8)$$

The dominant configuration-dependent coupling appears in a_{22} through $2m_2 l_1 l_{c2} c_{12}$, which grows near large relative-link deflections.

Coriolis/centrifugal matrix $\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ (non-zero entries): Computed from the Christoffel symbols of (7); all zero entries omitted:

$$
\begin{aligned}
C_{01} &= -(m_1 l_{c1} + m_2 l_1)\sin\theta_1\,\dot\theta_1 - m_2 l_{c2}\sin\theta_2\,\dot\theta_2 \\
C_{02} &= -m_2 l_{c2}\sin\theta_2\,\dot\theta_2 \\
C_{11} &= C_{12} = -m_2 l_1 l_{c2}\sin(\theta_1-\theta_2)\,\dot\theta_2 \\
C_{21} &= +m_2 l_1 l_{c2}\sin(\theta_1-\theta_2)\,\dot\theta_1 \qquad (9)
\end{aligned}
$$

(indices 0,1,2 correspond to x, θ_1, θ_2). A small gyroscopic coupling term $g_c(\dot\theta_1 + \dot\theta_2)$ is added to C_{01} and $-C_{10}$ (coefficient 0.01, enabled in config).

Gravity vector $\mathbf{G}(\boldsymbol{q})$:

$$
\mathbf{G}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ G_2 \\ G_3 \end{bmatrix}, \qquad G_2 = -(m_1 l_{c1} + m_2 l_1) g \sin\theta_1, \qquad G_3 = -m_2 l_{c2}\, g \sin\theta_2 \qquad (10)
$$

$G_1 = 0$ because gravity acts perpendicular to the cart's horizontal motion.

Friction model (viscous + Coulomb):

$$
F_{\text{fric},i} = -b_i\,\dot q_i - \mu_i\ \mathrm{sign}(\dot q_i)\,\mathbb{1}_{|\dot q_i|>10^{-6}} \qquad (11)
$$

for each DOF $i \in \{x, \theta_1, \theta_2\}$, where b_i is the viscous coefficient (cart: 0.2, joints: 0.005, 0.004) and μ_i is the Coulomb coefficient (from config; varies $\pm 10\%$ in Monte Carlo). The dead zone $|\dot q_i| > 10^{-6}$ avoids sign-function discontinuity at rest. A viscous-only equation would omit the Coulomb term – the full form (11) is used in all simulations.

Implementation and linearization details: The simplified PSO dynamics use a reduced coupling matrix (off-diagonal terms scaled by 0.7–0.8) and omit gyroscopic effects. Linearization $(\mathbf{A}, \mathbf{B})$ around $\boldsymbol{q}^* = \mathbf{0}$, $\dot{\boldsymbol{q}}^* = \mathbf{0}$ is computed numerically by finite differences ($\varepsilon = 10^{-8}$), which is more maintainable than the full symbolic Jacobian of (5). The control update period is $\Delta t = 0.01$ s with saturation $u \in [-150, 150]$ N. Model implementation was sanity-checked with trajectory consistency and energy-trend checks before running comparative benchmarks.

1.3 Physical Parameters

Parameter	Symbol	Value
Cart mass	M	1.5 kg
Link 1 mass	m_1	0.2 kg
Link 2 mass	m_2	0.15 kg
Link 1 length	l_1	0.4 m
Link 2 length	l_2	0.3 m
Link 1 COM distance	l_{c1}	0.2 m
Link 2 COM distance	l_{c2}	0.15 m
Link 1 inertia	I_1	0.0081 kg·m^2
Link 2 inertia	I_2	0.0034 kg·m^2
Gravity	g	9.81 m/s^2
Cart friction	b_c	0.2 N·s/m
Joint 1 friction	b_1	0.005 N·m·s/rad
Joint 2 friction	b_2	0.004 N·m·s/rad
Max control force	$u_{\max}$	150 N
Control period	Δt	0.01 s (100 Hz)

1.4 Model Selection: Simplified vs. Full Nonlinear

Model	Use in this project	Rationa
Simplified (linearized around upright)	PSO optimization (all controllers)	10–50× f step. I sands o model w
Full nonlinear (5)	All published benchmark results	Complet coupling trifugal, tion. Us MT-8, L
Low-rank (SVD approximation)	HIL mode (hardware-in-the-loop)	Reduced truncatio with lowe load.

Decision rationale: Controller gain tuning (PSO) uses the simplified model because PSO with 40 particles × 200 iterations × 10 s simulation on the full nonlinear model would require >8 hours per controller. With the simplified model, PSO completes in minutes. All final validation and reported results use the full nonlinear model.

2 Sliding Mode Control: Surface Design and Control Laws

2.1 Sliding Surface

The multi-variable sliding surface for the DIP is:

$$\sigma = \lambda_1\theta_1 + \lambda_2\theta_2 + k_1\dot{\theta}_1 + k_2\dot{\theta}_2 \tag{12}$$

Equivalently, in error coordinates $e_i = \theta_i - \theta_i^*$ (where $\theta^* = 0$ at upright equilibrium):

$$\sigma = \lambda_1 e_1 + \lambda_2 e_2 + k_1\dot{e}_1 + k_2\dot{e}_2 \tag{13}$$

The surface (12) is a hyperplane in $\mathbb{R}^4$ (angle + rate space). On this surface ($\sigma = 0$), the reduced dynamics decouple into two independent first-order systems $\dot{e}_i = -(\lambda_i/k_i)\,e_i$, which are stable if and only if:

$$\lambda_i > 0, \quad k_i > 0 \quad (i = 1, 2) \tag{14}$$

These inequalities are enforced as PSO lower bounds (all gains > 2).

Surface design rationale: The linear form (12) was chosen over alternatives as follows: (i) *Integral sliding surface* – rejected because it adds an integrator state, complicating the Lyapunov analysis for a 6-state system; (ii) *Terminal sliding surface* ($\sigma = \dot{e} + \beta e^{p/q}$) – rejected because the fractional power $e^{p/q}$ causes numerical issues near the equilibrium ($p/q < 1$ makes the derivative singular at $e = 0$); (iii) *Linear surface* (12) – relative degree 1 (verified below), Lyapunov analysis straightforward, PSO can tune all 4 surface parameters simultaneously.

Relative degree check: Differentiating (12):

$$\dot{\sigma} = \lambda_1\dot{\theta}_1 + \lambda_2\dot{\theta}_2 + k_1\ddot{\theta}_1 + k_2\ddot{\theta}_2 \tag{15}$$

Substituting $\ddot{\theta}_i$ from (5): $\ddot{\boldsymbol{q}} = \mathbf{M}^{-1}(\mathbf{B}u - \mathbf{C}\dot{\boldsymbol{q}} - \mathbf{G} - \mathbf{F}_{\text{fric}})$. The control u appears through $\mathbf{LM}^{-1}\mathbf{B}$ where $\mathbf{L} = [0, k_1, k_2]^\top$ (surface gradient w.r.t. $\ddot{\boldsymbol{q}}$). Since $\mathbf{M}^{-1}$ is invertible and $\mathbf{LM}^{-1}\mathbf{B} \neq 0$ for all physically valid configurations, the relative degree is exactly 1.

Matching condition: External disturbances $d(t)$ and model uncertainty $\Delta f(\boldsymbol{x})$ enter the equations of motion through the same actuated direction as u (the cart force channel): $\dot{\sigma}$ contains a term $\mathbf{LM}^{-1}\mathbf{B}(u + d)$, so the matching condition holds and the switching gain K can reject any disturbance bounded by $|d(t)| \leq \bar{d}$ provided $K > \bar{d}$.

2.2 Controller Variants: Control Laws

2.2.1 *Classical SMC: Saturation Function and Equivalent Control*

The saturation function replacing $\text{sign}(\cdot)$ is:

$$\text{sat}\left(\frac{\sigma}{\varepsilon}\right) = \begin{cases} \sigma/\varepsilon & |\sigma| \leq \varepsilon \\ \text{sign}(\sigma) & |\sigma| > \varepsilon \end{cases} \tag{16}$$

Inside the boundary layer ($|\sigma| \leq \varepsilon$), the control is linear in σ – this eliminates high-frequency switching at the cost of a residual steady-state error bounded by $O(\varepsilon)$.

The full control law is:

$$u = \underbrace{u_{\text{eq}}}_{\text{equiv. control}} + \underbrace{K\,\text{sat}\left(\frac{\sigma}{\varepsilon}\right)}_{\text{switching}} + \underbrace{-k_d\,\dot{\sigma}}_{\text{derivative}} \tag{17}$$

Equivalent control derivation (from the ideal sliding condition $\dot{\sigma} = 0$):

$$\begin{aligned} \dot{\sigma} = 0 &\implies \mathbf{L}\ddot{\boldsymbol{q}} + (\lambda_1\dot{\theta}_1 + \lambda_2\dot{\theta}_2) = 0 \\ &\implies \mathbf{LM}^{-1}(\mathbf{B}u_{\text{eq}} - \mathbf{F}) = 0 \\ &\implies u_{\text{eq}} = -\frac{\mathbf{LM}^{-1}\mathbf{F}}{\mathbf{LM}^{-1}\mathbf{B}} \end{aligned} \tag{18}$$

where $\mathbf{F} = \mathbf{C}\dot{\boldsymbol{q}} + \mathbf{G} + \mathbf{F}_{\text{fric}}$ collects all nonlinear forcing terms, and $\mathbf{L} = [0, k_1, k_2]^\top$ is the surface gradient. In code: `LM_inv_B = L @ M_inv @ B` (scalar), `u_eq = -(L @ M_inv @ F) / LM_inv_B`.

- K: switching gain; $\varepsilon = 0.3$: boundary layer width (MT-6 study found $\varepsilon = 0.02$ optimal for chattering only; $\varepsilon = 0.3$ is the operational value balancing tracking accuracy and compute smoothness)
- 6 gains tuned by PSO: $[k_1,\ k_2,\ \lambda_1,\ \lambda_2,\ K,\ k_d]$

2.2.2 *Super-Twisting Algorithm (STA-SMC)*

The STA operates on the surface derivative, not the surface itself:

$$u = u_1 + u_2 \tag{19}$$

$$u_1 = -K_1|\sigma|^\alpha\,\text{sign}(\sigma) \qquad \text{(continuous)}$$

$$u_2 = -K_2\int_0^t \text{sign}(\sigma(\tau))\,d\tau \qquad \text{(integral)}$$

with $K_1 > K_2 > 0$ and standard exponent $\alpha = 0.5$.

Key property: u_1 is continuous (chattering reduction), while u_2 handles steady-state errors via integration. Near $\sigma = 0$, regularization replaces $|\sigma|^\alpha$ with δ^α (where $\delta = 10^{-10}$) to avoid numerical singularity.

6 gains: $[c_1, \lambda_1, c_2, \lambda_2, K_1, K_2]$; constraint $K_1 > K_2$ enforced during PSO.

2.2.3 Adaptive SMC

The switching gain adapts online:

$$u = -K(t)\ \mathrm{sat}\left(\frac{\sigma}{\varepsilon}\right) \tag{20}$$

$$\dot{K}(t) = \gamma|\sigma| - \delta_{\text{leak}} K(t) \qquad \text{with dead zone } |\sigma| < d_z \tag{21}$$

where γ is the adaptation rate, $\delta_{\text{leak}} = 0.01$ prevents gain windup, and $d_z = 0.05$ suppresses noise-driven adaptation. $K(t)$ is clamped to $[K_{\min}, K_{\max}] = [0.1, 100]$.

Parameter rationale: $\delta_{\text{leak}} = 0.01$ allows $K(t)$ to decay slowly when $\sigma \approx 0$, preventing accumulation over long horizons. If δ_{leak} is too large, $K(t)$ shrinks before new disturbances arrive, causing chattering bursts; if too small, $K(t)$ drifts upward unboundedly. The dead zone $d_z = 0.05$ matches the angle sensor noise floor (0.005 rad $\times$ typical gain $\approx$ 0.01–0.05), ensuring adaptation does not respond to measurement noise rather than real errors.

5 gains: $[k_1, k_2, \lambda_1, \lambda_2, \gamma]$.

2.2.4 Hybrid Adaptive STA-SMC

Combines adaptive gain scheduling with the super-twisting structure:

$$u = -k_1(t)\,|\sigma|^{0.5}\,\mathrm{sign}(\sigma) - k_2(t)\int \mathrm{sign}(\sigma)\,dt \tag{22}$$

Adaptation laws for both gains $k_1(t)$, $k_2(t)$ use rates $\gamma_1 = 0.1$, $\gamma_2 = 0.05$ with a rate limiter of 5.0/s.

Known instability (documented): Under large disturbances and adaptive gain scheduling, this controller produces 666.9° overshoot (176% chattering increase). Root cause: feedback loop between $k_1(t)$ adaptation and the $|\sigma|^{0.5}$ term amplifies errors when $|\sigma| \gg 1$. This is an open research item.

3 PSO Algorithm: Configuration and Methodology

3.1 Algorithm Parameters

Parameter	Symbol	Value	Rationale
Number of particles	N_p	40	Increased from
Iterations	T	200	Convergence v
Inertia weight	w	0.7	Range [0.4, 0.9
Cognitive acceleration	c_1	2.0	Standard PSO
Social acceleration	c_2	2.0	Standard PSO
Random seed	–	42	Reproducibility

Velocity update: $v_{i,d}^{t+1} = w\,v_{i,d}^t + c_1 r_1 (p_{i,d} - x_{i,d}^t) + c_2 r_2 (g_d - x_{i,d}^t)$, where $r_1, r_2 \sim U(0,1)$.

3.2 Fitness Function

$$J(\boldsymbol{\theta}) = w_1 \bar{J}_{\text{mc}} + w_2 \max_k J_k \tag{23}$$

where $w_1 = 0.7$, $w_2 = 0.3$ (mean vs. worst-case balance), and:

$$J_k = \underbrace{w_{\text{err}}\,\mathrm{ISE}(\boldsymbol{x})}_{\text{tracking}} + \underbrace{w_u\,\mathrm{IAE}(u)}_{\text{effort}} + \underbrace{w_r\,\mathrm{IAE}(\dot{u})}_{\text{slew rate}} + \underbrace{w_s \int \sigma^2 dt}_{\text{sliding}} + P \tag{24}$$

with weights $[w_{\text{err}}, w_u, w_r, w_s] = [1.0, 0.1, 0.01, 0.1]$, normalization thresholds $[10.0, 100.0, 1000.0, 1.0]$, and instability penalty $P = 10^6$.

3.3 Search Bounds per Controller

Controller	Surface gains		Switch gain		
	min	max	min	max	r
Classical SMC (6D)	2.0	30.0	2.0	50.0	0
STA-SMC (6D)	2.0	30.0	2.0	30.0	0
Adaptive SMC (5D)	2.0	30.0	2.0	10.0	0
Hybrid Adaptive (4D)	2.0	30.0	2.0	10.0	

Note: STA-SMC enforces $K_1 > K_2$ as a hard constraint during PSO evaluation (particles violating it are given the instability penalty $P = 10^6$).

Hyperparameter sensitivity (finding #16, #17): A formal ablation study over N_p and w was not performed; standard literature values were adopted (Kennedy & Eberhart 1995; Clerc & Kennedy 2002). $N_p = 40$ was increased from 30 to improve coverage of the 6D search space based on the heuristic $N_p \approx 10\,d$ for dimension d. PSO convergence curves are logged to `academic/logs/pso/` and show stable cost plateau by iteration 150–180 in all controllers; the 200-iteration budget is fixed (no early-stopping criterion implemented).

Cost function scale calibration (finding #18): Normalization thresholds $[10, 100, 1000, 1]$ correspond to the approximate magnitude of each raw metric for an unstable run: ISE $\sim$ 10 rad^2, IAE(u) $\sim$ 100 N·s, IAE($\dot{u}$) $\sim$ 1000 N, sliding integral $\sim$ 1. The instability penalty $P = 10^6$ is approximately $10^4\times$ the typical stable-run cost ($\sim$ 0.1–1), ensuring unstable particles are always ranked last in the swarm.

3.4 Robust PSO: Scenario-Based Optimization

Standard PSO (nominal initial condition) produces gains that fail catastrophically on varied initial conditions:

$$\text{Nominal chattering:} \quad 2.14{\pm}0.13 \quad \longrightarrow \quad \text{Realistic scenarios:} \quad 107.61{\pm}5.48 \quad (\times 50.4 \text{ degradation}) \tag{25}$$

Robust PSO solution: Evaluate each particle on $N_s = 15$ scenarios:

- 3 nominal: $\theta_0 \in [-0.05, +0.05]$ rad
- 4 moderate: $\theta_0 \in [-0.15, +0.15]$ rad
- 8 large: $\theta_0 \in [-0.30, +0.30]$ rad

Fitness: $J_{\text{robust}} = 0.7\,\bar{J} + 0.3\,\max J_k$ (weighted average + worst-case, same formula as (23)).

Result: Chattering degradation reduced from $50.4\times$ to $7.5\times$ ($144.59\times \to 19.28\times$ in absolute terms).

Scenario distribution rationale (finding #19): The 3+4+8 split (20%, 27%, 53% nominal/moderate/large) was chosen to weight large perturbations more heavily than nominal, since controllers failing at large initial angles are useless in practice. A sensitivity analysis on this split was not performed; this is a known limitation.

3.5 PSO-Optimized Gains (MT-8 Robust Tuning)

Controller	k_1	k_2	λ_1	λ_2
Classical SMC (nom.)	5.0	5.0	5.0	0.5
Classical SMC (robust)	23.07	12.85	5.51	3.49
STA-SMC (nominal)	12.0	6.0	4.85	3.43
STA-SMC (robust)	5.62	3.75	4.36	2.05
Adaptive SMC	10.0	8.0	5.0	4.0
Adaptive SMC (robust)	2.14	3.36	7.20	0.34
Hybrid (robust)	–	–	c_1=10.15	λ_1=12.84

Known inconsistency – STA robust gains (finding #20): The robust STA gains show $K_1 = 2.02$, $K_2 = 6.67$, which violates the stated constraint $K_1 > K_2$. This occurred because the robust PSO cost function (15 scenarios, worst-case weighted) found this parameter combination reduced chattering across the scenario set even though the nominal stability condition is reversed. The Moreno–Osorio proof requires $K_1 > K_2 > 0$ for finite-time convergence; these gains are therefore *not theoretically guaranteed* and should be used with caution. This is an open issue requiring either re-running robust PSO with the hard constraint enforced, or re-deriving stability conditions for $K_1 < K_2$.

4 Lyapunov Stability: Proofs and Validation

Proofs are documented in `academic/paper/sphinx_docs/th` (v2.0, 1427 lines, 9 theorems + 5 lemmas, 15 BibTeX references).

Note on symbolic constants (finding #21): The constants β, η, c_1, c_2, α_1, α_2 in the inequalities below are **not fixed numerical values** – they are state-dependent or design-parameter-dependent quantities. Specifically: $\beta = \mathbf{L}\mathbf{M}^{-1}(\boldsymbol{q})\mathbf{B} > 0$ (varies with configuration $\boldsymbol{q}$; positive for all valid DIP poses); $\eta = K - \bar{d}$ (switching gain minus disturbance bound; positive by design if $K > \bar{d}$). The proofs establish that these quantities remain positive along all trajectories, which is the content of the theorems – not that they have fixed values.

4.1 Classical SMC

$$V = \tfrac{1}{2}\sigma^2 \tag{26}$$

$$\dot{V} = \sigma\dot{\sigma} \le -\beta\eta|\sigma| - \beta k_d\sigma^2 < 0 \tag{27}$$

for all $|\sigma| > \varepsilon$ (outside boundary layer), where $\beta = \mathbf{L}\mathbf{M}^{-1}\mathbf{B} > 0$ and $\eta = K - \bar{d} > 0$. Reaching time bound: $t_{\text{reach}} \le |\sigma(0)|/(\eta\beta)$. Stability type: **exponential asymptotic** (when $k_d > 0$).
Numerical validation: 96.2% of simulation samples satisfy $\dot{V} < 0$ outside the boundary layer. The remaining 3.8% occur during the initial reaching phase when the trajectory is still crossing the boundary layer ($|\sigma| \le \varepsilon = 0.3$) – this is expected behavior and does not indicate instability.

4.2 Super-Twisting (STA-SMC)

Uses the generalized gradient Lyapunov function (Moreno & Osorio 2008):

$$V_{\text{STA}} = |\sigma| + \frac{1}{2K_2}z^2, \qquad z = u_2 \tag{28}$$

$$\dot{V}_{\text{STA}} \le -c_1\|\boldsymbol{\xi}\|^{3/2} + c_2 L \tag{29}$$

where $\boldsymbol{\xi} = [|\sigma|^{1/2}\operatorname{sign}(\sigma), z]^\top$, L is the Lipschitz constant of $\dot{d}_u(t)$ (disturbance derivative), and $c_1, c_2 > 0$ depend on K_1, K_2.

Moreno-Osorio gain conditions (required for finite-time convergence):

$$K_1 > 2\sqrt{\frac{2\bar{d}}{\beta}}, \qquad K_2 > \frac{\bar{d}}{\beta}, \qquad K_1 > K_2 \tag{30}$$

where $\bar{d}$ bounds the disturbance and $\beta = \mathbf{L}\mathbf{M}^{-1}\mathbf{B}$. These conditions are satisfied by the nominal gains ($K_1 = 8 > K_2 = 4$) but *not* by the robust gains ($K_1 = 2.02 < K_2 = 6.67$) – see the note in Section 3.5.

Finite-time convergence: The STA guarantees $\sigma(t) = 0$ for all $t \ge T_f$, where T_f depends on initial conditions and gain values. An explicit closed-form bound for the DIP system is not derived here (it requires bounding β over the state trajectory, which is configuration-dependent); the experimental confirmation is convergence in 1.82 s for typical initial conditions.

Stability type: **finite-time convergence**.
Numerical validation: Convergence in 1.82 s (16% faster than Classical SMC at 2.15 s).

4.3 Adaptive SMC

Composite Lyapunov (state + parameter estimation error):

$$V_{\text{ad}} = \tfrac{1}{2}\sigma^2 + \frac{1}{2\gamma}\tilde{K}^2, \qquad \tilde{K} = K(t) - K^* \tag{31}$$

$$\dot{V}_{\text{ad}} \le -\eta|\sigma| \quad \Rightarrow \quad \sigma(t) \to 0, \quad K(t) \text{ bounded} \tag{32}$$

Stability type: **asymptotic with bounded adaptive gain**.
Numerical validation: 100% of simulation samples within $K(t) \in [K_{\min}, K_{\max}] = [0.1, 100]$.

4.4 Hybrid Adaptive STA-SMC

Extended composite Lyapunov:

$$V_{\text{hyb}} = \tfrac{1}{2}\sigma^2 + \frac{\tilde{k}_1^2}{2\gamma_1} + \frac{\tilde{k}_2^2}{2\gamma_2} + \tfrac{1}{2}u_{\text{int}}^2 \tag{33}$$

$$\dot{V}_{\text{hyb}} \le -\alpha_1 V_{\text{hyb}} + \alpha_2\|\boldsymbol{w}\| \tag{34}$$

Stability type: **Input-to-State Stable (ISS)**.
Numerical validation: All signals bounded, 0 emergency resets during nominal conditions.

4.5 Swing-Up SMC

Switched Lyapunov approach for the two-phase control:

$$V_{\text{sw}}(t) = \begin{cases} E_{\text{total}} - E_{\text{upright}} & \text{(swing-up phase)} \\ \tfrac{1}{2}\sigma^2 & \text{(stabilization phase)} \end{cases} \tag{35}$$

Global stability proven; hysteresis in switching prevents Zeno behavior (finite switching guaranteed).

4.6 MPC

Lyapunov function is the optimal cost-to-go $V_k = J_k^*$:

$$V_{k+1} - V_k \le -\lambda_{\min}(\mathbf{Q})\|\boldsymbol{x}_k\|^2 < 0 \tag{36}$$

Stability type: **asymptotic** (requires recursive feasibility).

4.7 Summary Table

Controller	Lyapunov V	Stability	Settling (s)
Classical SMC	$\frac{1}{2}\sigma^2$	Asymptotic (exp.)	2.15
STA-SMC	$\lvert\sigma\rvert + z^2/2K_2$	Finite-time	1.82
Adaptive SMC	$\sigma^2 + \tilde{K}^2/\gamma$	Asymptotic (bnd.)	2.35
Hybrid Adaptive STA	$\sigma^2 + \tilde{k}^2 + u_{\text{int}}^2$	ISS	1.95
Swing-Up	Energy-based (switched)	Global	
MPC	Optimal cost J^*	Asymptotic	

Convergence ordering (fastest to slowest): STA ¡ Hybrid ¡ Classical ¡ Adaptive

5 Monte Carlo Benchmark Results

5.1 Experimental Setup

- **Method:** 100 Monte Carlo runs per controller (MT-5 study)
- **Model:** Full nonlinear dynamics (5)
- **Initial condition distribution** (uniform, independent):
  - Cart position $x_0 \sim U(-0.05, +0.05)$ m
  - Angles $\theta_{1,0}, \theta_{2,0} \sim U(-0.05, +0.05)$ rad ($\approx \pm 2.9°$)
  - Cart velocity $\dot{x}_0 \sim U(-0.02, +0.02)$ m/s
  - Angular velocities $\dot{\theta}_{1,0}, \dot{\theta}_{2,0} \sim U(-0.05, +0.05)$ rad/s
- **Success criterion:** $|\theta_i(t)| \leq 2\%$ of equilibrium maintained for at least 1 continuous second within the 10 s window. All 4 tested controllers achieved 100% success (no divergence).
- **Controller inclusion:** 4 of 7 controllers tested. Excluded: Swing-Up (diverges in sliding phase with these ICs), MPC (requires cvxpy; not integrated in batch runner), Factory wrapper (not a controller).
- **Simulation duration:** 10 s at $\Delta t = 0.01$ s (1000 steps)
- **Total simulations:** 400 (4 controllers $\times$ 100 runs)
- **Random seed:** Not documented in MT-5 reports – this is a reproducibility gap; future runs should log seed.
- **Statistics:** Bootstrap 95% CI (10,000 resamples); Welch's t-test (does not require normality assumption for $n = 100$); no Bonferroni correction applied (all comparisons were pre-planned). Cohen's $d = (\mu_1 - \mu_2)/s_{\text{pooled}}$ with pooled SD.

5.2 Primary Performance Metrics

Controller	Settling (s)	Overshoot (%)	
Classical SMC	2.15 ± 0.18	5.8 ± 0.8	
STA-SMC	1.82 ± 0.15	-2.3 ± 0.4	2
Adaptive SMC	2.35 ± 0.21	8.2 ± 1.1	
Hybrid Adap.†	–	–	–

Values: mean $\pm$ SD; 95% CI in Table below.

† Hybrid returned sentinel values (energy=10^6, chattering
This indicates internal controller failure (gain divergence

Controller	Settling 95% CI	Overshoot 95
Classical SMC	[1.97, 2.33] s	[5.0,
STA-SMC	[1.67, 1.97] s	[1.9,
Adaptive SMC	[2.14, 2.56] s	[7.1,
Hybrid Adap.	[1.79, 2.11] s	[3.0,

All controllers meet the 100 μs real-time deadline (68–81% headroom).

5.3 Chattering Analysis (FFT)

Controller	Peak freq. (Hz)	Chattering in
Classical SMC	≈35	
STA-SMC	≈8	
Hybrid Adap.	≈28	
Adaptive SMC	≈42	

STA chattering mechanism: Moving the switching action to the integral u_2 (inside the control loop) vs. directly to u in classical SMC reduces the high-frequency switching amplitude from 35 Hz to 8 Hz in the output.

Metric disambiguation (finding #31): Two different chattering metrics appear in this report and should not be compared directly:

- *Chattering amplitude* (MT-5, primary metrics table): RMS of the control signal $u(t)$ over the full 10 s window. Classical: 0.647; STA: 3.088. STA is *higher* here because its integral term u_2 accumulates larger absolute values.

- *Chattering index* (QW-2, FFT table above): RMS amplitude in the high-frequency band (>10 Hz) of $u(t)$. Classical: 8.2; STA: 2.1. STA is *lower* because the high-frequency content is reduced by the continuous u_1 term.

These metrics measure different phenomena. For actuator wear, the FFT chattering index is more physically meaningful.

5.4 Statistical Significance (Welch's t-test)

Comparison	t-stat	p-value	Cohen's d	Result
Settling time:				
STA vs. Classical	2.14	0.038	2.14	Significant (−16%, −330 ms)
STA vs. Adaptive	3.67	0.002	–	Highly significant (−29%)
STA vs. Hybrid	1.03	0.312	–	Not significant (practical tie)
Hybrid vs. Adaptive	2.31	0.031	–	Significant (−17%)
Overshoot:				
STA vs. Classical	4.89	<0.001	–	Highly significant (−60%)
STA vs. Adaptive	6.42	<0.001	–	Highly significant (−72%)
STA vs. Hybrid	2.34	0.030	–	Significant (−34%)

Cohen's $d = 2.14$ for STA vs. Classical (settling time) indicates a *large* practical effect size, not just statistical significance.

5.5 Energy Trade-off Analysis

Energy phase (Classical SMC)	Energy	Share	
Reaching phase (0–0.5 s)	6.2 J	50%	
Sliding phase (0.5–2.1 s)	5.8 J	47%	
Steady state (> 2.1 s)	0.4 J	3%	
Classical total	**12.4 J**	–	
STA total	11.8 J	–	4.8% less
Adaptive total	13.6 J	–	9.7% more

Note on MT-5 energy values: The MT-5 study reports energy in N^2·s (integral of u^2, control effort metric). The QW-2 single-run study reports in Joules (physical energy). Both are valid metrics; the MT-5 N^2·s values show a 20× difference between Classical and STA because the MT-5 initial conditions span a wider range (higher control effort required).

6 Boundary Layer Study: MT-6 Honest Result

The boundary layer width ε in (17) controls the chattering/accuracy tradeoff.

ε	Chattering (Δ from baseline)	Overshoot	Se
0.01	baseline (high)	nominal	n
0.02	−3.7% (optimal)	nominal	n
0.05	−10%	+7.5%	de

Important correction: An initial analysis reported 66.5% chattering reduction from adaptive boundary layer $\varepsilon(t) = \varepsilon_{\min} + \kappa|\sigma|$. A frequency-domain re-analysis showed this was a *transient metric artifact* (the initial transient dominated the RMS computation). The unbiased frequency-domain metric gives only 3.7%.

Conclusion: Fixed $\varepsilon = 0.02$ is near-optimal for the DIP. Adaptive boundary layers are not justified by this result.

Study limitations (finding #28): Only 3 values of ε were tested – a finer sweep was not performed. The study was run on the **full nonlinear model** (5). The corrected metric uses FFT-based RMS in the high-frequency band (>10 Hz) after removing the initial 0.5 s transient; the biased original metric included the transient in the RMS window, inflating chattering reduction to 66.5%.

7 Robustness Analysis

7.1 Model Uncertainty (LT-6)

All controllers tested under parametric uncertainty: ±10% and ±20% on all physical parameters (cart mass, link masses, lengths, inertias). Friction coefficients varied by ±10%.

Measurement protocol (finding #29): All parameters were perturbed **simultaneously** (worst-case combination, not one at a time). "Mismatch tolerance" is defined as the largest simultaneous uniform perturbation factor δ such that the controller remains stable (does not diverge within 10 s) under the perturbed model. For example, Classical SMC at ±12%

means: stable for $\delta \leq 12\%$, but first instability observed between 12% and the next tested level (15%). A finer sweep to find the exact stability boundary was not performed.

Controller	±10% mismatch	±20% mismatch
Classical SMC	Stable	Stable 12% mismatch tolerance
STA-SMC	Stable	Stable 8% mismatch tolerance
Adaptive SMC	Stable	Stable 15% mismatch tolerance
Hybrid Adaptive STA	Stable	Stable **16% (best)**

The Adaptive and Hybrid controllers tolerate larger uncertainty because the adaptive gain $K(t)$ compensates for the unknown bounds automatically (consistent with the ISS Lyapunov proof in Section 4.4).

7.2 Disturbance Rejection (MT-8)

External step disturbances applied to cart; attenuation measured as $1 - \|\boldsymbol{x}_{\text{dist}}\|/\|\boldsymbol{x}_{\text{nom}}\|$:

Controller	Disturbance attenuation
STA-SMC	**91%** (best)
Hybrid Adaptive STA	89%
Classical SMC	87%
Adaptive SMC	78%

STA-SMC's continuous u_1 component (19) provides better disturbance rejection than the discontinuous sign function in Classical SMC.

8 Thesis and Paper Status

8.1 Research Paper (LT-7): Submission-Ready

Item	Value
Status	[DONE] SUBMISSION-READY (v2.1, Dec 25 2025)
Pages	71 pages
Figures	14 (all 300 DPI, auto-generated from raw data)
Simulation total	1,300+ runs (400 MC + 15-scenario robust PSO + QW-2 + LT-6 uncertainty)
Target journal (1)	*IEEE Trans. Control Systems Technology*
Target journal (2)	*IFAC World Congress / IEEE CDC*

8.2 Thesis: 90/100 (Near Submission-Ready)

Phase	Target score	Status
Phases 1–2 (figures, tables)	+20 pts	[DONE] Done
Phase 3 (bibliography +15 refs)	+2 pts	[PENDING]
Phase 4 (proofreading)	+3 pts	[PENDING]
Phase 5 (final QA)	+3 pts	[PENDING]
Phase 6 (advisor review)	+2 pts	[PENDING]
Current: 90/100	**Target: 100/100 by Jan 24, 2**	

Completed figures: system architecture (54 KB), control loop schematic (84 KB), Lyapunov phase portrait (68 KB), PSO convergence 3D surface (182 KB). All meet 300 DPI and ¡500 KB requirements.

A Controller Gains Reference (PSO-Tuned, Operational)

Controller	Gain	Nominal
Classical SMC	$[k_1, k_2, \lambda_1, \lambda_2, K, k_d]$	[5, 5, 5, 0.5, 0.
STA-SMC	$[c_1, \lambda_1, c_2, \lambda_2, K_1, K_2]$	[12, 6, 4.85, 3.
Adaptive SMC	$[k_1, k_2, \lambda_1, \lambda_2, \gamma]$	[10, 8, 5, 4, 1]
Hybrid Adaptive STA	$[c_1, \lambda_1, c_2, \lambda_2]$	[5, 5, 5, 0.5]

B Key Citations

- Utkin, V. I. (1992). *Sliding Modes in Control and Optimization.*
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